北京郵電大學

Queen Mary University of London

For examiners' use only

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EBU6501 A

Joint Programme Examinations 2021/22

EBU6501 Middleware

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Ethan Lau, Dr Gokop Goteng

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave

Question 1

a) In the Publish and Subscribe model, ***topic-based*** and ***content-based*** are the common forms of the message filtering systems for publishing, processing, and reception. Describe the functionalities of the two filtering systems with aids of diagrams.

[6 marks]

	Do not write in this column
	6 marks

b) There are TWO methods to ***receive*** a message from a ***Topic*** for the JMS ***Message Consumer***, namely the ***synchronous*** and ***asynchronous***. Figure 1 below shows the JMS programming API where the ***Connection*** and ***Session*** objects have been created. The JMS ***Connection Factory*** and ***Destination*** in this case has as well been configured administratively.

[8 marks]

```
1.// create a Connection & client ID
2.TopicConnection connection = connectionFactory.createTopicConnection();
3.connection.setClientID(clientId);
4.
5.// create a Session
6.TopicSession session = connection.createTopicSession(false,
7.Session.AUTO_ACKNOWLEDGE);
```

Figure 1

i) Briefly explain the main difference between the **synchronous** and **asynchronous** type of the JMS messaging consumption.

(2 marks)

ii) In your own words, explain the function of the **`session`** in the JMS Message consumer model.

(2 marks)

iii) From Figure 1, write the JMS code that starts the connection right after the **`session`** is created.

(1 mark)

iv) Then, use **`TopicSubscriber`** interface to create a receiver for **`MyTopic.`**

(1 mark)

v) Finally, implement the **`Message Listener`** interface that acts as event handler for registering and listening for messages (you do NOT need to include the **`onMessage()`** method).

(2 marks)

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	8 marks

c) Figure 2 below shows the directory structure of the Tomcat container.

[4 marks]

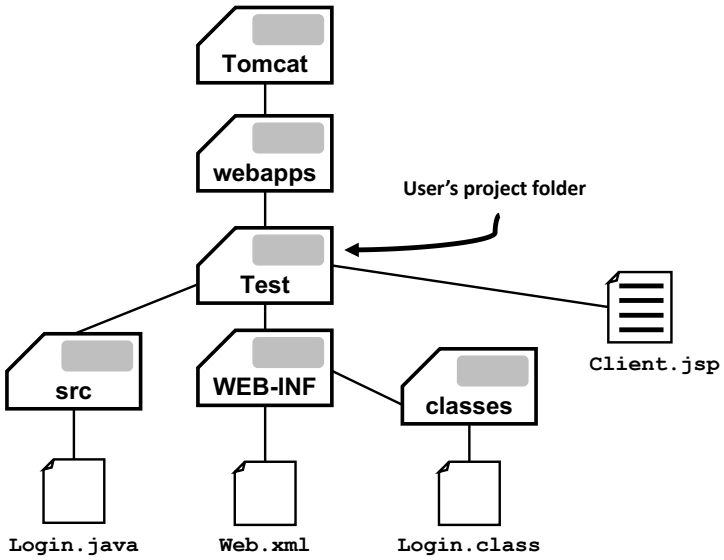


Figure 2

i) Provide the correct ***URL*** using HTTP protocol and appropriate port number to access the web page "**`client.jsp`**" if you are running the system installed on your local machine.

(2 marks)

ii) Define the term given to the file "**`web.xml`**" and explain why it is located under the **WEB-INF** directory**.**

(2 marks)

	Do not write in this column
	4 marks

d) Figure 3a presents the life cycle method of a servlet that hosts applications by means of a request-response programming model. Then, Figure 3b presents the Java code implementation of the request-response programming model. A servlet class **`MyServlet`** is created that extends **`HttpServlet`** class. The code however is missing the implementation of the ***Service*** and ***Destroy*** methods. The servlet shall return an HTML page that display the current value of the field **`myMsg`** as the text in an H1 element. Complete the following:

[7 marks]

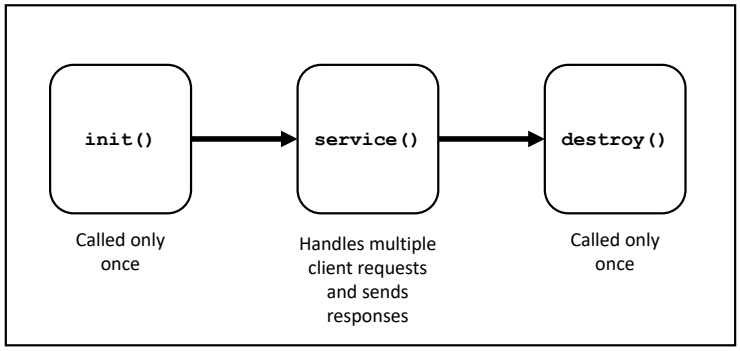


Figure 3a

```
public class MyServlet extends HttpServlet {

   private String mymsg;
   // init() method
   public void init() throws ServletException {
      mymsg = "Hello Servlet!";
   }

   // ### Missing the service() method

   // ### Missing the destroy() method
}
```

Figure 3b

i) Write the required **`service()`** method that called by the server to allow a servlet to handle a GET request.

(2 marks)

ii) Set the content type as "text/html" for the webpage.

(1 mark)

iii) Write message **`myMsg`** to the webpage.

(2 marks)

iv) Close the stream for garbage collection and destroy the servlet.

(2 marks)

	Do not write in this column

Question 2

a) In Spring Framework, Inversion of Control (IoC) is a principle in software engineering by which the control of objects or portions of a program is transferred to a container or framework as shown in Figure 4. It is most often used in the context of object-oriented programming such as Java.

[11 marks]

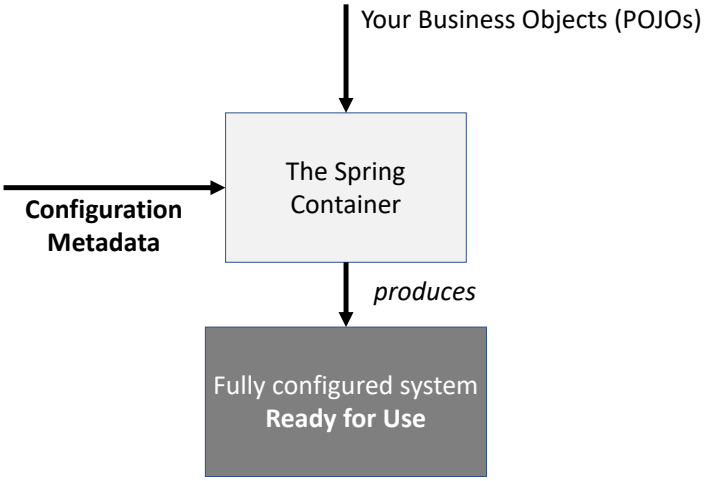


Figure 4

i) Based on the information in Figure 4, explain how the container in Spring allows the system to be fully configured and is ready for the application deployment and provide TWO examples of the configuration metadata.

(4 marks)

ii) A bean XML definition "**`module.xml`**" and the Java class "**`moduleBean.java`**" using the Setter Dependency Injection are shown in Figure 5 with the bean ID = "**`moduleBean`**" and the property name = "**`subject`**". Now that the **`moduleBean`** is defined in both the XML definition and the Java class, write the **`main()`** method of class **`MyModuleMain`** as the main ***Application Context*** to load the configured bean.

(7 marks)

```
//module.xml
<bean id="StudentBean"  class="org.module.moduleBean">
      <property name="module" value="Middleware"/>
</bean>

//moduleBean.java
package org.module;
public class moduleBean {
      String module;
      public String getModule()
            { return module; }
      public void setModule(String module)
            { this.module = module; }
}
```

Figure 5

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b) This question is about threads.

[14 marks]

i) Explain how threads are synchronizing in the Java concurrent programming using the ***Mutual exclusion*** and ***Cooperation*** mechanisms.

(2 marks)

ii) Figure 6 below is a Java implementation of Threads in a Java concurrency programming. Explain what the code is doing in **lines 3**, **11**, **24** and **32**.

(8 marks)

iii) Predict and write the outputs of the Java program in Figure 6.

(4 marks)

```
1.public class ThreadA {
2. public static void main(String[] args){
3.   ThreadB b = new ThreadB();
4.   System.out.println("Main thread initialising...");
5.     synchronized(b){
6.       b.start();
7.       System.out.println("Thread B started.");
8.       try{
9.          System.out.println("Waiting for Thread B to complete...");
10.         while(!b.finished) {
11.         b.wait();}
12.       }catch(InterruptedException e){
13.        e.printStackTrace();}
14.      System.out.println("Total is: " + b.total);
15.    }
16.  }
17.}
18.
19.  Public class ThreadB extends Thread{
20.    int total=0;
21.    boolean finished;
22.    @Override
23.    public void run(){
24.      synchronized(this){
25.        try {
26.          for(int i=0; i<5 ; i++){
27.            total += i;
28.          }
29.        }
30.        finally {
31.          finished = true;
32.          this.notify();
33.        }
34.      }
35.    System.out.println("Thread completed.");
36.   }
37.  }
```

Figure 6

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Question 3

a) This question is on JavaScript programming:

[12 marks]

i) A weather forecasting station displays every day's temperature in Celsius, but its main algorithm for calculating the temperature uses Fahrenheit. Write a JavaScript function that will convert Fahrenheit to Celsius and prints the result. Given that Fahrenheit is equal to ((Fahrenheit-32) × 5/9) Celsius, use the **`var`** key word to initialise Fahrenheit to 100. Use **`<script>`** to start and end your program to keep it simple.

(5 marks)

ii) Describe and differentiate the JavaScript statements under **Statements A** and **Statements B** below:

(7 marks)

Statements A

```
let x = "John Wang";
let x = 0;
```

Statements B

```
var x = "John Wang";
var x = 0;
```

Figure 7

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	12 marks

b) This question is about web security:

[10 marks]

i) Describe how ***Public Key Cryptography*** works to ensure secure sharing of files among computer users.

(4 marks)

ii) How do the **DDoS** and **Cross Site Request Forgery (CSRF)** security attack on computer networks?

(**6 marks**)

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		10 marks

c) A medical equipment manufacturing company has designed and developed some medical devices to test and diagnose COVID-19 patients remotely using Internet of Things (IoT) framework. However, the devices have constraints in energy and data processing power, and they cannot use the normal computer protocols such as TCP/IP, etc. Suggest any THREE IoT protocols you think the company may try to use to accommodate these constraints.

[3 marks]

	Do not write in this column
	3 marks

Question marking $= \frac{}{12} + \frac{}{10} + \frac{}{3} = \frac{}{25}$

Question 4

a) Using your knowledge of the Open Services Gateway Initiatives (OSGi) architecture learnt, describe the functions of ***Modules***, ***Service Registry*** and ***Java Virtual Machine (JVM)*** in OSGi.

[6 marks]

	Do not write in this column
	6 marks

b) This question is on Serverless Applications and Microservices:

[8 marks]

i) Describe the concept of ***Serverless Architectures*** and ***Microservices*** in software development and discuss the role that ***Spring Cloud*** plays in implementation of Microservices in distributed computing systems.

(6 marks)

ii) Think and provide TWO examples of ***AWS Serverless Applications*** used in the development of distributed applications.

(**2 marks**)

	Do not write in this column

	8 marks

c) Write a simple Java program with a class called **`URLClass`** that uses the Java URL class and BufferedReader class to connect, read and print from the Oracle URL given as "**https://www.oracle.com**".

[11 marks]

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	11 marks

Question marking $= \frac{}{6} + \frac{}{8} + \frac{}{11} = \frac{}{25}$

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2021-2022
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Page 18 of 20

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2021-2022

Rough Working

Page 19 of 20

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2021-2022

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Page 20 of 20